

Engine Stage Planner

User Manual

A mod for Kerbal Space Program 1.12.x

Engine Stage Planner helps you pick engines. In the VAB/SPH it tells you which engines will actually meet a delta-v and TWR target for a given stage. In flight it plots telemetry on a scrolling graph.

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1. Installation

1.1 Required mods

Engine Stage Planner will not run without these. Install them first.

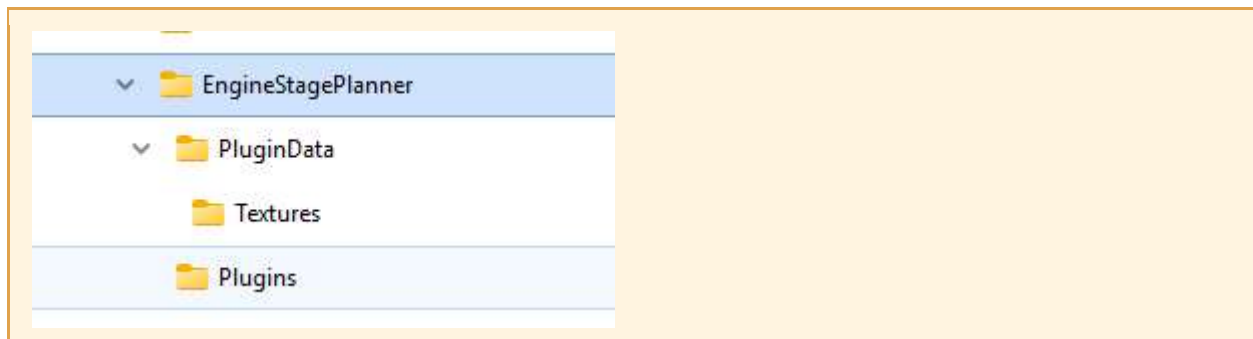
Mod	File it must provide
ToolbarController	GameData/001_ToolbarControl/Plugins/ToolbarControl.dll
ClickThroughBlocker	GameData/000_ClickThroughBlocker/Plugins/ClickThroughBlocker.dll

1.2 Copy the files

Place these four files under GameData/EngineStagePlanner:

From	To
EngineStagePlanner/bin/Release/EngineStagePlanner.dll	GameData/EngineStagePlanner/Plugins/EngineStagePlanner.dll
EngineStagePlanner/PluginData/Textures/icon_38.png	GameData/EngineStagePlanner/PluginData/Textures/icon_38.png
EngineStagePlanner/PluginData/Textures/icon_24.png	GameData/EngineStagePlanner/PluginData/Textures/icon_24.png
EngineStagePlanner.version	GameData/EngineStagePlanner/EngineStagePlanner.version

Start KSP. If the toolbar button appears in the VAB, you're installed.



2. Opening the window

One toolbar button does everything. It appears in the VAB, the SPH, and in Flight.

- **In the editor**, it opens the planner.
- **In flight**, it opens the telemetry graph.

The planner window starts **closed** every time you enter the editor — click the button to open it.

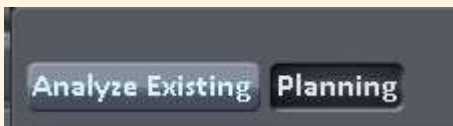


The Engine Stage Planner button on the KSP application toolbar.

Both windows can be dragged from any empty area and resized using the grip in the lower-right corner. Clicks won't leak through to parts behind the window, and the windows close themselves automatically when KSP starts loading a new scene.

3. The two editor modes

Mode	Use it when
Analyze Existing	You already have a craft built and want to know which engine to put on a stage that already has tanks and fuel.
Planning	You have a delta-v target and a payload mass, and you want the mod to size the stage for you from scratch.



The Analyze Existing / Planning selector at the top of the planner.

4. Using Analyze Existing

Step 1 — Pick the stage

You have three ways to choose which stage to analyze:

1. Type a number into the **Stage** field.
2. Click **Pick Stage**, then click any part on your craft. The planner jumps to that part's stage.
3. Click the orange header of any stage in KSP's normal staging list on the right of the screen.

While Pick Stage is armed, the mod temporarily locks the editor so your click selects a part instead of grabbing and detaching it. Normal editing returns as soon as you release the mouse.

In this mode you cannot select a stage number higher than the craft's current maximum stage — the field clamps itself, including when you add or remove stages.



Pick Stage armed, cursor hovering a fuel tank on the craft.

Step 2 — Set up the simulation environment

The left column splits into two panes. The top pane, **Simulation environment**, is always visible and holds everything that affects the run:

- **Minimum TWR** — candidates below this are rejected.
- **TWR gravity** — filled in automatically from the body you select.
- **Max engines** — the largest cluster size the mod will try.
- **Simulation filters** — engine name text filter, bulkhead matching, ignore monopropellant.
- **Simulate all engines** — run the sweep.
- **Simulation status** — progress and result count.

Below it, the **Existing stage** pane shows what the mod found on your craft: wet and dry masses, payload above the stage, propellant currently loaded versus full capacity, current engine mass, burn time, tank volume, and the list of engines already installed.

Existing stage			
Craft wet mass			31.35 t
Craft dry mass			8.06 t
Payload above stage			0 t
Stage propellant (current)			19.25 t
Stage propellant (full)			19.25 t
KSP stage wet mass			31.256 t
KSP stage dry mass			7.966 t
KSP stage fuel mass			23.29 t
KSP current engine mass			4.5 t
KSP vehicle start mass			31.256 t
KSP vehicle end mass			18.383 t
KSP current burn			23.7 s
Stage tank capacity			3100 units
Known tank volume			15500 L
Current engines			
Engine	ASL kN	Vac kN	Isp ASL/Vac
LV-T45	167.969	215	250/320
"Swivel" Liquid			

The two stacked panes — Simulation environment above Existing stage.

Step 3 — Choose a body and altitude

Click the current-body button to open a vertical list of celestial bodies. Pick one, then set an altitude with the slider. The mod uses KSP's own atmospheric pressure model and each engine's real atmosphere curve, so the atmospheric numbers are what you'd actually get there.

The body list draws on top of the rest of the window rather than pushing it down, and it stays anchored to the button as you move the planner around.



The body list open beneath the current-body button.

Step 4 — Read the results

The candidate list shows every compatible engine, simulated against the propellant **actually loaded** in that stage right now — not maximum tank capacity. See Section 6 for what each column means.

Filters available in this mode

- **Engine name** — live, case-insensitive text search.
- **Match stage bulkhead size** — only shows engines whose top attach node size matches the top node of the engines already on that stage.
- **Ignore monopropellant** — hides any candidate whose mass-bearing propellant includes MonoPropellant.

5. Using Planning

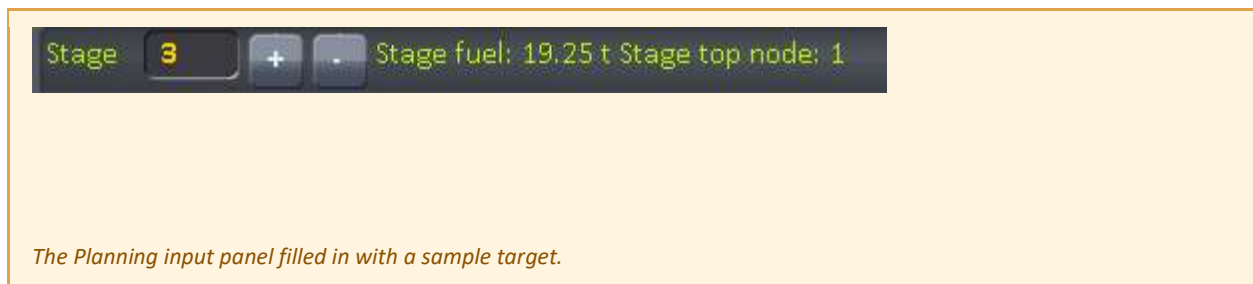
Planning sizes a stage that doesn't exist yet.

Step 1 — Enter your targets

Input	Meaning
Target delta-v	Interpreted as vacuum delta-v
Minimum TWR	Rejection threshold

Input	Meaning
Payload	Mass sitting on top of this stage
Planet / body	Sets TWR gravity automatically
Altitude	Only affects the atmospheric columns
Max engine count	Largest cluster to try
Tank structural ratio	See Step 2

You may type a **future stage number** here that's higher than your craft's current maximum — that's the point of the mode. Planning's top row is deliberately simpler than Analyze Existing's; it has no Pick Stage, Craft max, or Craft wet controls.



Step 2 — Set the tank ratio

Planning has to guess tank dry mass before you've picked a tank, so it uses a **tank dry mass ÷ propellant mass** ratio. 0.125 is a reasonable stock-like default. If you select a stage that already has resource-bearing parts, the mod infers a ratio from them for you.

The solver loops on this: propellant mass depends on tank dry mass, and tank dry mass depends on propellant mass.

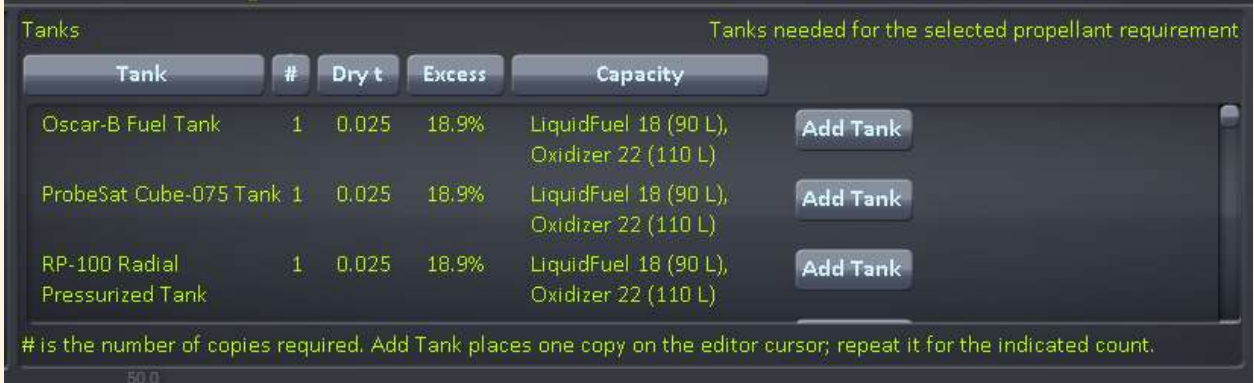
Step 3 — Read the results and the tank table

Because Planning sizes the stage from **vacuum** delta-v, the stage geometry is fixed. Moving the altitude slider will **not** change propellant load, wet/dry mass, vacuum delta-v, tank requirements, or burn time. It only changes the atmospheric delta-v, atmospheric thrust, and atmospheric TWR shown for that fixed stage.

Below the engine results, the **Tanks needed** table lists each available tank type with:

- how many copies you'd need
- total tank dry mass

- excess capacity
- supplied resource capacity



The Tanks needed table showing several tank options.

6. Reading the candidate table

The list is capped at 300 pixels tall and scrolls, so growing the window won't let the engine list swallow the UI. Sorting and filter changes recalculate immediately.

Column	What it is
Engine	Part name
#	How many engines in this candidate configuration
Stage Wet t	Mass of this stage only — no payload, no upper stages
Start Mass t	Full vehicle mass accelerated at ignition; this is what drives delta-v and TWR
Atm Δv	Delta-v at your selected body and altitude
Vac Δv	Vacuum delta-v
ASL kN/eng	Sea-level thrust of one engine
Vac kN/eng	Vacuum thrust of one engine
TWR	Atmospheric TWR, using total thrust of all engines

Column	What it is
Max TWR	Vacuum-thrust TWR, sortable
Isp	Specific impulse
Burn	Burn duration
Fuel	Propellant required
Δv /Fund	Atmospheric delta-v divided by total engine cost
Add	Puts that engine on the editor cursor

Note the deliberate split: the per-engine kN columns show **one** engine, while TWR uses **all** of them. The selected-result detail panel shows both per-engine and total thrust so you can see the multiplication.

Candidates (click a column heading to sort; click engine name for details; Add selects it for editor placement)

Cost Efficiency = atmospheric Δv at the selected planet/altitude per Fund of engine cost.

Engine	#	Stage Wet t	Atm Δv	Vac Δv	ASL kN/eng	Vac kN/eng	TWR	Max TWR	Isp ASL/Vac	Burn	
LV-1R "Spider" Liquid Fuel Engine	1	0.068	2241	2500	1.793	2	2.68	2.99	260/290	57s	Add
24-77 "Twitch" Liquid Fuel Engine	1	0.273	2371	2500	15.172	16	5.67	5.98	275/290	28s	Add
DTE Streamline S-12 Liquid Fuel Engine	1	0.303	1984	2500	9.524	12	3.20	4.03	250/315	43s	Add
48-7S "Spark" Liquid Fuel Engine	1	0.386	2070	2500	16.563	20	4.37	5.28	265/320	33s	Add
O-10 "Puff" MonoPropellant Fuel Engine	1	0.395	1200	2500	9.6	20	2.48	5.16	120/250	31s	Add
RV-1 "Cub" Vernier Engine	1	0.558	2258	2500	28.903	32	5.28	5.85	280/310	30s	Add
DTE Streamline S-30 "Sphere" Liquid Fuel Engine	1	0.634	2131	2500	25.574	30	4.11	4.83	260/305	36s	Add
DTE Streamline 2-120 "Duet" Liquid Fuel Engine	1	2.377	2109	2500	101.25	120	4.34	5.15	270/320	34s	Add
DTE Streamline 1-60-S "Solo" Liquid Fuel Engine	1	2.525	2109	2500	50.625	60	2.04	2.42	270/320	73s	Add
LV-T91 "Cheetah" Liquid Fuel Engine	1	2.624	1056	2500	52.817	125	2.05	4.86	150/355	37s	Add

The full candidate list with several engines ranked.

Sorting

Pick one optimization goal from the vertical toggle group:

- Lowest wet stage mass
- Lowest propellant
- Lowest engine cost
- Shortest burn
- Highest TWR
- Highest Isp

Propellant display

Requirements are shown in KSP resource units and in mass. Where the resource definition supplies a volume, estimated physical tank volume in litres is shown too.

7. Adding parts to your craft

Three buttons place parts, all using KSP's normal editor spawning path:

- **Add** on any candidate row — places that engine on the cursor.
- **Add Engine** on the selected result — same thing for the highlighted candidate.
- **Add Tank** in the Tanks needed table — places **one** copy. Click it again for each copy the table calls for.

For multimode engines, both listed modes point at the same underlying part.

Optionally, the planner can close itself after a successful placement. Analyze Existing and Planning have **separate** preferences for this, and the window only closes if KSP actually put the part on your cursor.

IMAGE PLACEHOLDER
images/add-engine.png

An engine attached to the cursor immediately after clicking Add.

8. Settings

Open **Settings** from the editor planner. Everything is saved to:

```
GameData/EngineStagePlanner/PluginData/EngineStagePlannerSettings.cfg
```

and restored next time you run KSP. The Settings window drags from any unused background area.



The Settings window with its page tabs visible.

Engine Columns page

Turn any candidate table column on or off individually — Engine, count, Stage Wet, Start Mass, atmospheric and vacuum delta-v, atmospheric and vacuum thrust, cost efficiency, TWR, Max TWR, Isp, Burn, Fuel, and the row Add button. **Show All** and **Hide All** are one-click presets.

Analyze Existing page

Controls which informational lines appear in the **Existing stage** pane: craft wet/dry mass, payload above stage, current and full stage propellant, KSP stage wet/dry/fuel masses, current engine mass, vehicle start/end mass, current burn, tank capacity and known tank volume, the Current Engines section, and stage-resource lines.

Simulation controls are never hidden by these settings. This page also holds the close-planner-after-Add option for Analyze Existing.

Planning page

Its own independent close-planner-after-Add / Add Engine / Add Tank option.

Appearance page

Use solid backgrounds for all editor windows gives the planner and the editor Settings window an opaque dark underlay while keeping normal KSP borders and title styling. Off by default. It does not affect flight windows.

9. Flight telemetry plotter

Click the same toolbar button in flight to open the graph.



Starting a recording

- **Start at Launch** can be armed while the vessel is in PRELAUNCH. When the vessel leaves PRELAUNCH, existing data is cleared and recording begins automatically. Once you're already flying, this button is disabled.
- **Reset** clears the current plot samples and stage markers.
- Sampling pauses with the game and resumes only after you unpause and the sample delay has elapsed.

Sample interval

The **Sample** field sits at the top of the plotter toolbar. Default is **0.25 seconds**. The – and + buttons step it by 0.25 s, and you can type an exact value from 0.05 to 60 seconds. It sets the minimum real-time delay before the next sample is recorded.

What can be plotted

- Surface velocity and orbital velocity
- Vertical speed
- Acceleration
- G-force
- Altitude ASL and altitude above surface
- Dynamic pressure (q)
- Vehicle mass
- Angle of attack
- Every resource carried by the active vessel
- Output from any stock ModuleEnviroSensor on the vessel

All available sources are recorded while plotting, so switching which series are visible can redraw data you already collected.

Flight settings

Open **Settings** from the graph to choose series. Flight data, resources, and sensors are grouped separately, each with All/None controls. The table shows **Plot**, **Sensor**, **Max**, and **Units**. The main plotted list also shows **Current** and **Max** per series.



The flight Settings series selection table.

How the graph scales

- The graph fills left to right with fixed horizontal sample spacing. Once it hits the right edge, the window scrolls and old data moves left.
- Elapsed time since plotting started is labelled along the bottom as m:ss, or h:mm:ss on long runs.
- Stage activations appear as orange dashed vertical lines labelled Stage N. They scroll with the data and are cleared by Reset or Start at Launch.
- **Altitude** series use a configurable **Altitude chart top** with a zero baseline. By default that's the launch body's atmosphereDepth, or 10% of body diameter for an airless body. Altitude intentionally stores no Max value because this setting controls its scale.
- **Every other series** scales to a per-vessel historical maximum recorded across all launches and reverts in the current save. The lower bound follows currently visible data, so series with wildly different units can share one graph.

Maxima persist in:

```
GameData/EngineStagePlanner/PluginData/FlightSensorMaxima.tsv
```

Reset deliberately preserves them. Use **Clear Max Values for Vessel** to wipe the current craft's non-altitude maxima without clearing the current plot.

Exporting

Export CSV writes the selected series as comma-separated text. **Export PNG** writes an image of the currently visible graph. Both land in Screenshots/EngineStagePlanner.

10. How the numbers are calculated

You don't need this section to use the mod, but it explains why the numbers are what they are.

Which engines appear as candidates

Candidates come from KSP's loaded part database **after** the editor's ExcludeFilters are applied. If another mod hides an engine through KSP's editor filter API, it's hidden here too. The currently selected editor category is not applied.

Thrust and TWR

- TWR gravity comes from the selected body's surface gravity automatically.
- For conventional rockets, vacuum thrust is the configured ModuleEngines.maxThrust; atmospheric thrust follows the ratio of atmospheric to vacuum Isp from the engine's atmosphere curve.
- Engines whose fuel flow genuinely depends on atmosphere or velocity use KSP's own flow calculation, with a sanity fallback.
- Analyze Existing honours each installed engine's editor thrust limiter and shows individual ASL and vacuum thrust per engine.
- Analyze Existing also carries the mass of loaded stage resources the candidate engine cannot burn, so they drag down TWR and delta-v exactly as they would in game.

Mass boundaries

Analyze Existing prefers KSP's own VesselDeltaV / DeltaVStageInfo results, which respect the stock staging and fuel-flow model:

- **Stage Wet t** uses stageMass
- **Start Mass t** uses the full vehicle startMass
- Retained payload is startMass – stageMass

Installed engine mass is removed using DeltaVPartInfo.dryMass before candidate engine mass is added, so the existing engine isn't counted twice.

The branch scanner still identifies the stage's resource inventory, tank volume, and engine list, and acts as a fallback when the stock delta-v simulation isn't ready yet. Parts found by the branch scan are excluded from the payload-above-stage sum so they aren't double counted.

Burn time

Mass flow is derived from vacuum thrust and vacuum Isp rather than getMaxFuelFlow:

$$\dot{m} = F / (I_{sp} \times g_0)$$

$$t = m_{\text{Propellant}} \times I_{\text{sp}} \times g_0 / F$$

Burn rate is therefore independent of the selected body and altitude. In Analyze Existing, `mPropellant` is the mass of the propellant mixture the candidate engine can actually burn, limited by what's loaded in the stage — KSP's whole-stage `fuelMass` is not substituted in. This stops unrelated stage resources from inflating burn duration.

The rocket equation uses KSP's standard gravity constant, 9.80665 m/s².

Resources

The full propellant list is read from each `ModuleEngines` / `ModuleEnginesFX` module. Nothing about `LiquidFuel/Oxidizer` or any other mixture is assumed.

- Every resource with a positive engine ratio is kept in the requirement set, including intake and electrical resources.
- Resources that contribute to `Isp` and have mass drive the rocket equation.
- `ignoreForIsp` resources stay visible as auxiliary requirements but don't define mass flow.
- Physical volume comes from `PartResourceDefinition.volume`. There is no hard-coded volume table, so stock and modded resources both work from whatever definitions your install has loaded.

11. Known limits

The planner is a good heuristic, not a full flight simulator. These are not currently modelled:

- Dynamic tank type/configuration switching by `RealFuels`, B9, MFT-style modules after the prefab loads.
- Jet velocity curves and intake-air flight-state modelling beyond static pressure-based `Isp` and thrust.
- Engine variants controlled by arbitrary third-party modules beyond loaded `ModuleEngines`.
- Full `MechJeb`-style fuel-flow simulation for asparagus, crossfeed, and drop-tank arrangements.

For unusual modded fuel-flow systems, use **Planning** with a manually entered payload mass — it doesn't depend on the stage scan being right.

12. Version notes

Version	Change
0.2	Tanks needed table; Add / Add Engine / Add Tank editor placement; bulkhead-size filtering; Ignore monopropellant
0.3.6	Existing-stage simulation uses current resource amounts, not max capacity; Planning fixed to vacuum Δv so altitude no longer resizes the stage
0.3.7	Candidate list capped at 300 px and scrollable; immediate filter recalculation; Δv /Fund column
0.4.12	Sample interval moved to the flight toolbar with ± 0.25 s buttons; stage markers driven by GameEvents.onStageActivate
0.4.13	Default flight sample interval returned to 0.25 s
0.5.0	Generic engine-resource support; density and volume read from PartResourceDefinition
0.5.7	Stock startMass/endMass boundaries; burn time from thrust and Isp instead of getMaxFuelFlow; elapsed-time labels on flight graph; planner starts closed
0.5.13	Burn duration computed from propellant the candidate engine can actually use
0.5.14–0.5.19	Stage clamping in Analyze Existing; wider mass/thrust columns; column and info-line visibility settings; persistent draggable Settings; per-mode close-after-Add; Alt+P removed; Simulation environment split into its own pane; Appearance opaque-window option
0.5.25	Corrected the scene-switch callback signature to GameEvents.FromToAction<GameScenes, GameScenes>